

High-fidelity detection of large-scale atom arrays in an optical lattice

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(Dated: January 19, 2024)

I. SUPPLEMENTARY INFORMATION

A. Details on the Experimental Setup

The optical lattice in our experiment is created with a single 1040 nm beam retro-reflected in 4f-configuration on the xy -plane, see Fig. 1a. The beam crossing angle θ is chosen to be 26° resulting in a lattice spacing $a_x = 579(2)$ nm along the x -axis and $a_y = 1187(18)$ nm along the y -axis. The foci of all four beams overlap at the center of the field of view of the objective with waists of $20\text{ }\mu\text{m}$ ($100\text{ }\mu\text{m}$) along the z -axis (in the xy -plane). With the polarization of all beams aligned along z -axis, multiple-pass interference yields a trap frequency that can reach 150 kHz (300 kHz) along the x -axis (y -axis), at a trap depth of 1.47 mK with 12 W optical power. In the z -axis, the harmonic confinement of the lattice reaches about $2\pi \times 5.9$ kHz. The tweezer spacing in both axes is precisely chosen to be $6a_y$ and $3a_x$, respectively. Using 650 mW optical power at 520 nm, we create a 2D array of $21 \times 19 = 399$ tweezers that have trap depth $140\text{ }\mu\text{K}$ each. The waist of the tweezers is estimated to be $472(3)$ nm from independent calibrations of trap depth and trap frequency. The trap frequency at $140\text{ }\mu\text{K}$ tweezers depth is about $\omega_r/2\pi = 77.6(5)$ kHz radially as measured from sideband spectra. From tweezer waist and radial trap frequency, we deduce an axial trap frequency of $\omega_z/2\pi = 19.2(5)$ kHz. The trap depths for 399 tweezers are equalized to a root-mean-square intensity variation to 2.2% by minimizing variations in measured differential lightshifts on the $^1\text{S}_0$ - $^3\text{P}_1$ transition of an initial tweezer pattern pre-corrected for diffraction efficiency, see Fig S1. We carefully superimpose the tweezer array with the lattice by aligning their foci in all three dimensions. The global tweezer array position on the xy -plane is adjusted actively by feedback to a diffraction grating phase on the spatial light modulator to maintain optimal overlap with the slowly drifting lattice potential.

To initialize experiments involving tweezers, we first load a $140\text{ }\mu\text{K}$ -deep tweezer array with atoms from a single-frequency magneto-optical trap [1] on the 689 nm transition, followed by the parity projection in tweezers. The lattice is on during the loading state for loading enhancement. The filling fraction is typically 45%. Next, atoms are cooled to the radial ground state in the tweezers via sideband cooling in a magic magnetic field of 14 G oriented at an angle $\phi = 5.6^\circ$ relative to the linear tweezer polarization along the y -axis [2]. This leaves the atoms in the ground state along the radial directions with an estimated average ground-state population of $97^{+2}_{-5}\%$. For imaging characterization in the lattice, we then transfer the atoms from the tweezers to the optical lattice by ramping up the lattice trap depth and subsequently ramping down the tweezer trap depth in 5 ms.

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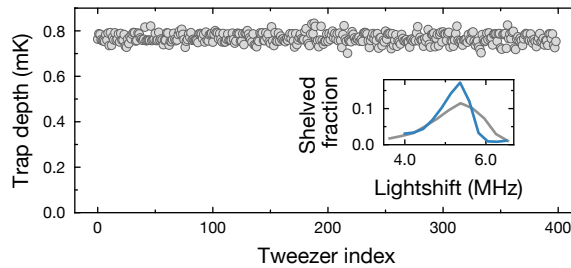


FIG. S1: **Tweezer-resolved trap depth after feedback.** With three rounds of equalization using lightshift measurements, the root-mean-square trap depth inhomogeneity reaches about 2.2% for 399 tweezers. The array-averaged $^1\text{S}_0$ - $^3\text{P}_{1,|m_J|=1}$ shelving spectra before (gray) and after corrections (blue) are shown in the inset. The linewidth narrowing shows a smaller spread of trap depths as a result of the homogenization.

B. Tweezer hologram generation

1. Tweezer generation

We generate our tweezer array holographically using a phase-only spatial-light modulator. The algorithm for creating the hologram largely follows earlier work [3] and involves a variant of the Gerchberg–Saxton algorithm where the phases of tweezers in the focal plane are kept stationary towards the end of the optimization. In the optical setup, we use a collimated beam with a waist of 4.2 mm incident on the SLM chip of dimension 12.8 mm $\times$ 15.9 mm. A telescope consisting of a 300 mm and 750 mm achromatic lens relays the imprinted phase pattern to the back focal plane of the high-resolution objective. The objective then transforms the phase pattern into an array of tightly-focused tweezers in the atomic plane.

Undiffracted light at the SLM is blocked in an intermediate imaging plane. An added blazing phase grating steers the diffracted beam by about 0.5° away from the undiffracted light and defines the new optical axis, with which we align our optics downstream. To avoid clipping at the input aperture of the objective, we impose a circular aperture of 11.5 mm in diameter on the SLM, such that the light fills less than 90% of the input pupil of the objective [5]. This aperture is assumed in all phase-retrieval algorithms. To correct wavefront distortion, we sample the SLM light shortly before it enters the objective and focus the beam onto a camera. Using phase-shifting interferometry [6, 7] with super-pixels on the SLM, we are able to suppress the RMS wavefront error to below $\lambda/14$.

2. Tweezer amplitude equalization

The first step to equalizing the individual depths of the optical tweezers consists of a compensation of the finite diffraction efficiency of the SLM, which can be understood as a consequence of the diffraction effect on the finite-sized SLM pixels. The diffraction efficiency $\eta(\theta_x, \theta_y)$ can be calibrated as a function of diffraction angle (θ_x, θ_y) using a diagnostic camera. The target tweezer amplitude is then weighted by $1/\sqrt{\eta}$ in the phase-retrieval algorithm [3]. As a second step, we use the atomic signal to spectroscopically probe the homogeneity of the tweezer array via the differential light shift induced on the cooling transition. At our tweezer wavelength of 520 nm, the $\sigma_{\pm}$ -component of the 689 nm transition exhibits about 7.3 MHz/mK differential lightshift, such that it is sensitive to spatial variations in tweezer depth. The spectroscopy is performed at a magnetic bias field of 14 G along the tweezer polarization axis to break the Zeeman degeneracy in the $^3\text{P}_1$ state. We adapt our optimization procedure for the two-step equalization protocol. The first step consists of 40 iterations of Gerchberg–Saxton algorithm (GSA), five iterations of weighted GSA, and another 35 iterations where the phase of each tweezer on the image plane is frozen. Tweezer arrays created at this point can typically reach less than about 8% intensity inhomogeneity for our 399 tweezer array. Subsequently, we spectroscopically probe the tweezer depth U_i , where i denotes the tweezer index, and adjust each tweezer amplitude by a factor $\sqrt{U_i}/\sqrt{\bar{U}}$, where $\sqrt{\bar{U}}$ is the value averaged over all i . After this second round of homogenization, the tweezer intensity inhomogeneity can typically reach below 2.2% after three iterations, see Fig S1.

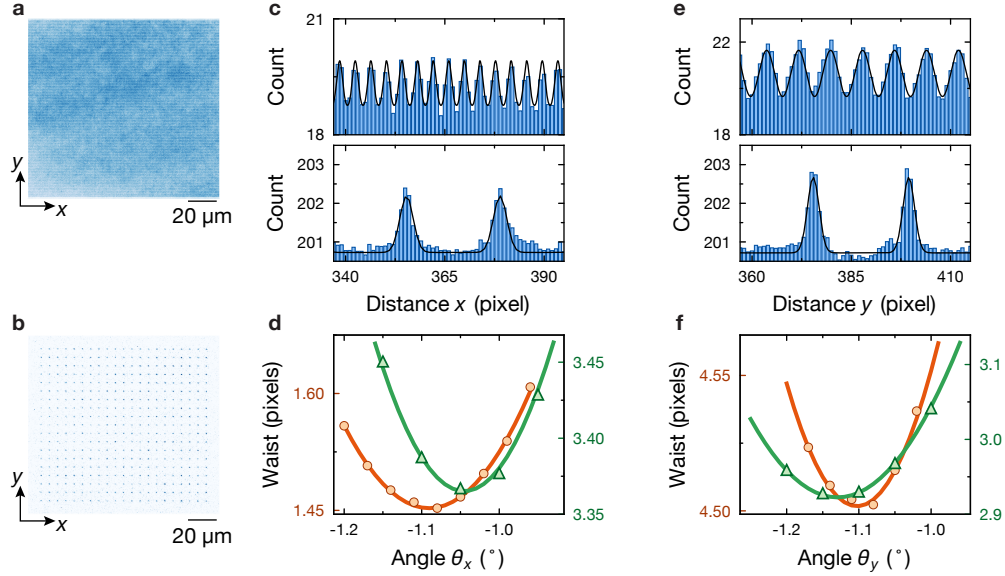


FIG. S2: **Determination of spacings and coordinate angles of tweezers and lattice.** **a** An averaged fluorescence image of a sparsely-loaded lattice used for the analysis. **b** An averaged fluorescence image of a tweezer array imaged in tweezers. For images of 2D fluorescence patterns defined by two orthogonal symmetry axes (like **a** and **b**), the 1D fluorescence count, which results from projecting the image array along some axis, exhibits smaller peak overlap when the projection axis coincides with either symmetry axis of the array [2, 4]. Such a case for $x(y)$ -axis is shown for lattice in the upper panel and the tweezer in the lower panel in **c(e)**. Any deviation from such a condition results in an increased peak overlap (**d, f**). For the analysis, the dependency of waist on the angle of projection axis is fitted with $\sigma_0 [1 + \beta \{1 - \cos(\gamma(\theta_i - \theta_{i,0}))\}]$ for $i = x, y$ [2]. In both **d** and **f**, lattice data (circle) and fit (solid line) are red, whereas tweezer data (triangles) and fit (solid line) are green. Once the minimum-waist angle is found, the lattice and tweezer array spacing and waist can be extracted by fitting a grid of Gaussians functions in the rotated frame of reference (**c, e**). In summary, we extract lattice coordinate angles $\theta_x^{\text{lat}}(\theta_y^{\text{lat}}) = -1.090(0)^\circ(-1.102(1)^\circ)$ and tweezer coordinate angles $\theta_x^{\text{tw}}(\theta_y^{\text{tw}}) = -1.0452(2)(-1.1296(1))^\circ$. For the spacings, we find lattice constants of $a_x(a_y) = 3.92(6)(8.04(9))$ pixels and tweezers spacing of $c_x^{\text{tw}}(c_y^{\text{tw}}) = 23.564(6)(24.047(6))$ pixels.

3. Determining spacings and angles of tweezers and lattice potentials

To create a tweezer array commensurate with the lattice, we need to determine the lattice coordinate system. For this, we largely follow the work of [2, 4], and base all analysis on an averaged fluorescence images of a sparsely-populated lattice loaded from the MOT, see Fig S2a. We eliminate a small mismatch in the rotation angles of the lattice and tweezers relative to the camera pixel grid by rotating the image, followed by a 1D averaging along one of the two directions. The alignment is optimal when the 1D averaged images display the highest contrast (or smallest fringe-width). Then, by fitting a grid of Gaussian profiles to the fluorescence peaks, we can identify the spacing. Such an analysis is done for both axes independently, yielding lattice spacings $a_x(a_y) = 3.92(6)(8.04(9))$ pixels and coordinate angles $\theta_x^{\text{lat}}(\theta_y^{\text{lat}}) = -1.090(0)^\circ(-1.102(1)^\circ)$, see Fig. S2. From the known wavelength of the lattice laser of 1040.139(1) nm in vacuum, we can deduce that one camera pixel is equal to 147.6(23) nm in the atomic plane. The lattice constants expressed in SI are then $a_x = 579(2)$ nm and $a_y = 1187(18)$ nm, respectively.

Once the coordinate system of the lattice is known, it is fairly straightforward to shape the tweezer array such that both potentials are commensurate. We define tweezer spacing c_i^{tw} such that $c_x^{\text{tw}} = 6a_x$ and $c_y^{\text{tw}} = 3a_y$, and $\theta_i^{\text{tw}} = \theta_i^{\text{lat}}$ for both x and y . With fluorescence imaging in tweezers only, we are able to verify the ratio of spacings directly. With a similar analysis as for the lattice, we find that the tweezer array has spacing $c_x^{\text{tw}}(c_y^{\text{tw}}) = 23.564(6)(24.047(6))$ pixels respectively and coordinate angles $\theta_x^{\text{tw}}(\theta_y^{\text{tw}}) = -1.0452(2)(-1.1296(1))^\circ$, see Fig S2.

C. Radial ground-state re-cooling after imaging in the lattice

Our tweezers have well-resolved radial sidebands and are therefore conducive to ground-state cooling. After atoms are imaged in the lattice and transferred back to tweezers, they in general acquire an elevated temperature, as indicated by the prominent cooling sideband in the sideband thermometry, see Fig S4a. We show that we can still perform

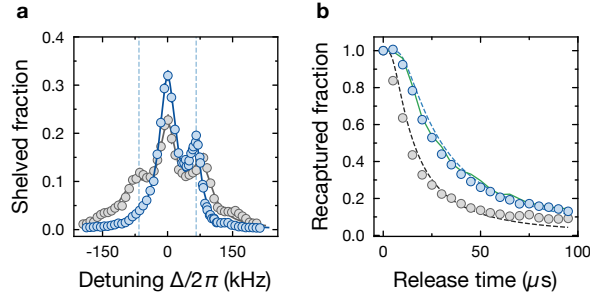


FIG. S3: **Reinitialization in radial motional ground-state after transfer.** **a** Radial sideband spectra of atoms loaded back into tweezers without (gray) and with (blue) sideband cooling after transfer. From the asymmetry of the spectrum, we extract ground-state fraction $98_{-3}^{+0}\%$ with and $37_{-17}^{+40}\%$ without sideband cooling after the transfer. **b** Release-and-recapture measurements in tweezers. We record the recovered fraction of atoms after a variable release time with (blue) and without (gray) sideband cooling after transfer. From a 3D Monte Carlo simulation, we extract temperatures of $5\ \mu\text{K}$ ($17\ \mu\text{K}$) with (without) sideband cooling after transfer, which is consistent with the temperature deduced from mean motional excitation measured by sideband spectroscopy. The fits are shown as blue and gray dashed lines respectively. The result of a release-and-recapture measurement before transferring to the lattice is indicated in green.

resolved sideband cooling on the radial degree of freedom and obtain atoms with $98_{-3}^{+0}\%$ ground-state fraction. This is comparable to the fraction we can achieve in tweezers by itself without involving the lattice at all. We cross-check the cooling with the release-and-recapture in tweezers, see Fig S4b. The result is consistent with the temperature estimated from sideband thermometry.

D. Laser cooling on narrow optical transitions

Narrow optical transitions found in alkaline-earth or alkaline-earth-like atomic species are well suited for a variety of cooling techniques that leverage differential trapping between the two states coupled via the narrow transition [8–13]. Generally, three conditions can be distinguished: (i) the trap frequency of the excited state ω_e is larger than that of the ground state, ω_g (“attractive Sisyphus regime”, $\omega_e > \omega_g$), (ii) the trap frequency of the excited state is smaller than that of the ground state (“repulsive Sisyphus regime”, $\omega_e < \omega_g$) and (iii) both the excited and ground state are equally trapped (“magic regime”, $\omega_e = \omega_g$). In the magic trapping regime, sideband cooling can be performed, where motional quanta are removed in a controlled way from the atoms by the light tuned to the resonance of the cooling sideband at a detuning $\Delta = -\omega_e$ from the free-space resonance at $\Delta = 0$. Here, equal and strongly confining trapping potentials for both ground and excited states are essential as this allows atoms to change their internal states without significant motional heating. However, even for unequal polarizabilities, one can find closed cycles of atoms undergoing excitation and deexcitation, during which the net kinetic energy can be reduced and converted to energy removed by emitted light. For attractive Sisyphus cooling, the cooling light is tuned to the *red* of the free-space resonance ($\Delta < 0$), and optimized to resonantly excite atoms residing at the trap center, whereas for repulsive Sisyphus cooling, the light is tuned to the *blue* of the free-space resonance ($\Delta > 0$), and optimized to resonantly excite atoms away from the trap center. Note that in the case of repulsive Sisyphus cooling, the light is still detuned to the red of the light-shifted resonance in the center of the trap, setting an effective “Sisyphus cap” in the detuning, above which an atom would be heated and repelled from the trap [9]. In consequence, whereas attractive Sisyphus cooling features an intrinsically stable cooling configuration and provides cooling independent of atom temperature, repulsive Sisyphus cooling can lead to runaway heating in case the temperature and hence potential energy of an atom exceeds the Sisyphus cap, see Fig 1e.

E. Estimating classification fidelity from histogram fitting

To measure the survival probability in imaging with low error, we need an accurate protocol of distinguishing the presence of an atom from the background. A common technique involves summing up fluorescence counts scattered over a region of interest (roi) centered at individual atoms and comparing the result against a threshold [9, 12]. To increase the signal-to-noise ratio in the detection, we first of all apply a de-convolution filter derived from the measured point-spread function to individual atomic fluorescence before summing up the counts in a square roi 3×3 lattice sites large. On account of our use of tweezers in loading the lattice, the position of the atoms in the lattice is susceptible to

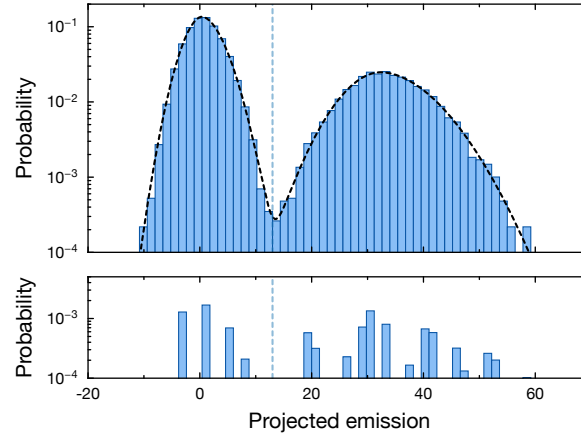


FIG. S4: **Classification fidelity estimate from histogram fitting** The fluorescence of individual atoms is de-convolved with a measured point-spread function and summed over a region of interest spanning 3×3 lattice sites. The resultant counts (in units proportional to photons) from 60 shots are then aggregated in a histogram (upper panel) on logarithmic scale to reveal low probability events. We fit the histogram with two Gaussian-like functions (black, dashed) and find a threshold (blue, dashed) that simultaneously minimize the false positive and false negative rate. The classification fidelity is then estimated to be 99.971(1)%. The fitting residual (lower panel) is small and indicates the accuracy of the model.

small variation from shot to shot, as a result of relative time-dependent phase drift and imperfect time-independent incommensurability between the two potentials. We heuristically find that identifying the location of atoms with a lattice sites-resolved occupation reconstruction aids in centering the roi. Consequently, the histogram exhibits a smooth profile that is amenable to analytical modeling, see Fig S4. We empirically fit the one-atom emission and background separately using a modified Gaussian function $P(x, x_0) = A \exp \left[-\frac{(x-x_0)^n}{\sigma^2} \right]$ where n is allowed to deviate from 2 and differs depending on whether $x < x_0$ or $x \geq x_0$. An optimal threshold can be found by minimizing the probability of detecting false positive and false negative. We find that the final classification fidelity is insensitive to the histogram bin size. The error bar in the classification is estimated from assigning the fitting residual to falsely detected events according to the worse case scenario.

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